

# Ujaan Das

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## EDUCATION

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| <b>University of Warwick</b><br><i>Master of Science in Computer Science (Predicted Distinction)</i>                                                                                                                                                                         | Coventry, United Kingdom<br>Sep. 2025 – Sep. 2026             |
| <b>Hong Kong University of Science and Technology</b><br><i>Bachelor of Engineering in Computer Engineering</i>                                                                                                                                                              | Hong Kong SAR<br>Sep. 2021 – May 2025                         |
| <b>Northwestern University</b><br><i>Exchange Semester</i>                                                                                                                                                                                                                   | Illinois, USA<br>Jan. 2024 – Apr. 2024                        |
| <b>West Island School</b><br><i>International Baccalaureate (41/45; HL: Computer Science, Physics, Psychology)</i><br><i>iGCSEs (English (A*), Mathematics (8), Sciences (9); Art (7), Business Studies (B), Computer Science (A), Spanish (A), Global Perspectives (A))</i> | Hong Kong SAR<br>Sep. 2019 – May 2021<br>Sep. 2017 – May 2019 |

## EXPERIENCE

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| <b>Stellerus Technology</b><br><i>Software Engineering Intern (Part Time)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Jan. 2025 – Aug. 2025<br>Hong Kong SAR |
| <ul style="list-style-type: none"><li>Re-designed a chatbot execution pipeline using a multi-step LLM workflow, implementing a custom command DSL for structured tool orchestration that reduced token overhead by 25% per session</li><li>Standardized message contracts and schema enforcement across the pipeline, eliminating 90% of parsing-related regressions and reducing runtime errors by 30%</li><li>Co-developed the SpaceGPT VLM platform awarded a Gold Medal at the 50th Geneva International Invention Exhibition; engineered the technical demo and automated deployment via Docker and Github Actions</li></ul>           |                                        |
| <b>Research Assistant</b><br><i>CASTLE Lab, HKUST</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | May 2024 – Sep. 2024<br>Hong Kong SAR  |
| <ul style="list-style-type: none"><li>Engineered high-throughput semantic-analysis pipelines in Python and Java to extract semantic signals from 30k+ scraped GitHub repositories to inform research in LLM-assisted testcase generation</li><li>Integrated Tree-sitter to parse multi-million-line codebases into ASTs and CFGs, converting complex source code into lightweight, queryable intermediate representations</li><li>Optimized this parsing logic to cut per-project analysis time by 45%, ultimately increasing usable data yield by 3.2x to support large-scale experiments, slated for publication in ACM ICSE'27</li></ul> |                                        |
| <b>Stellerus Technology</b><br><i>Software Engineering Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | May 2023 – Aug. 2023<br>Hong Kong SAR  |
| <ul style="list-style-type: none"><li>Developed RESTful APIs using FastAPI and MongoDB to orchestrate multi-user task queues for live satellite data, designed to sustain throughput of 8,000+ requests/sec with &lt;150ms latency and minimal queue times</li><li>Parallelized CPU-bottlenecked operations via a Celery/Redis worker pool, reducing processing time by 34%</li><li>Partnered with frontend team to containerize the SPA and backend using Docker and orchestrated automated CI/CD pipelines for Azure Container Apps, cutting deployment time by 20%</li></ul>                                                             |                                        |
| <b>HCI Initiative, HKUST</b><br><i>Research Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Jun. 2022 – Aug. 2022<br>Hong Kong SAR |
| <ul style="list-style-type: none"><li>Engineered a real-time telemetry and metrics pipeline processing 50K+ events/session, enabling analysis across 4K+ user interactions in large-scale experiments</li><li>Collaborated with post-docs to build a modular React + TypeScript frontend using custom hooks and Tailwind CSS, improving component reusability and streamlining development across multiple researchers</li><li>Contributed to FARPLS, a feature-augmented preference learning system that improved label quality and user engagement without increasing cognitive load (published at ACM/IUI'24)</li></ul>                  |                                        |

## PROJECTS

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| <b>MSc Dissertation - Invariants</b>   <i>C++, Gtest, CMake, Python</i>                                                                                                                                                                                                     | <a href="#">Github</a> |
| <ul style="list-style-type: none"><li>Applying formal systems theory to LLMs for improved structured output generation via runtime constraints</li><li>Architecting a C++ DSL and compiler with Python bindings for seamless integration into inference pipelines</li></ul> |                        |
| <b>Snip</b>   <i>C++, Termios, Gtest, CMake, Nix</i>                                                                                                                                                                                                                        | <a href="#">Github</a> |
| <ul style="list-style-type: none"><li>Actively maintaining a modal text editor built from ground-up with asynchronous, event-driven architecture</li></ul>                                                                                                                  |                        |
| <b>yumcha</b>   <i>Go, C, CGo, Cocoa, Nix</i>                                                                                                                                                                                                                               | <a href="#">Github</a> |

- Engineered a macOS tiling window manager in Go and C using Cocoa/CoreGraphics, implementing an i3-style tree layout engine for dynamic, scriptable window management without disabling SIP

**Undergraduate FYP - Follow-Me Robot** | *C++, Python, ROS2, Raspberry Pi, Docker*

[Github](#)

- Developed a real-time UWB tracking system via custom ROS2 Python/C++ nodes, applying multi-rate Kalman and EMA filtering for stable sub-centimeter localization under noisy conditions

**oojsite** | *Go, HTML, TailwindCSS, Nix*

[Github](#)

- Built a static site generator in Go to compile Markdown into composable HTML, actively used for [my blog](#)

**TerraViz** | *Next.js, React, TypeScript, CesiumJS, TailwindCSS*

[DevPost](#)

- Built an interactive geospatial platform in Next.js/React for real-time exploration of climate and land-use data

## SKILLS

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**Languages:** C++, Python, Go, C, TypeScript, SQL, JavaScript, HTML/CSS

**Frameworks & Libraries:** GTest, CMake, FastAPI, Next.js, React.js, Node.js, LangGraph, Pydantic

**Databases:** MongoDB, PostgreSQL, MySQL, SQLite, Redis

**Tools & Cloud:** Docker, Nix, Azure, AWS, GitHub Actions, NGINX, Make, Git